

Chapter: 01. Information Systems in Global Business Today

Learning Objectives

After reading this chapter, you will be able to answer the following questions:

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

What Is Management Information Systems?

Data

It streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use. Data refers to raw, unprocessed facts and figures that lack context or meaning. It can take various forms, such as numbers, text, images, and sound. Data on its own does not provide value until it is processed or organized. In an IT and IS context, data is the foundational building block for creating information and knowledge.

Information

Data that have been shaped into a form that is meaningful and useful to human beings. Information is the summarization of data. Technically, data are raw facts and figures that are processed into information, such as summaries and totals. Information is processed or organized data that has meaning and value. When raw data is analyzed, interpreted, and structured in a way that it becomes relevant and useful for decision making, it becomes information. For example, data like “1000” becomes information when it is contextualized as “1000 units sold” or “1000 dollars earned.”

Differences between Data and Information

Although the two terms are often used interchangeably, they are not the same thing. Data is the raw material, while information is the finished product that management actually uses. The table below summarizes the key differences.

Basis	Data	Information
Meaning	Raw, unorganized facts and figures	Processed data that has meaning and context
Form	Numbers, text, symbols, images, or sound with no structure	Organized as reports, summaries, charts, or statements
Usefulness	Not directly useful for decision making on its own	Directly useful for decision making and problem solving
Dependency	Exists independently, collected before processing	Depends on data, cannot exist without it
Example	"1000", "Dhaka", "March"	"1000 units were sold in Dhaka in March"

Information Technology

Information technology consists of all the hardware and software that a firm needs to use in order to achieve its business objectives. This includes not only computer machines, storage devices, and handheld mobile devices, but also software, such as the Windows or Linux operating systems, the Microsoft Office desktop productivity suite, and the many thousands of computer programs that can be found in a typical large firm. Information Technology (IT) refers to the use of computers, software, networks, and other digital systems to process, store, transmit, and retrieve data and information. IT encompasses a wide range of tools and techniques that enable organizations and individuals to manage and manipulate information efficiently. This includes hardware (such as computers, servers, and mobile devices), software (such as operating systems, applications, and databases), and networking infrastructure (such as the internet, intranets, and cloud computing). IT is integral to various industries, facilitating communication, automation, decision making, and business processes.

Information Systems

Information systems can be defined technically as a set of interrelated components that collect, process, store and distribute information to support decision making and control in organization. In addition to supporting decision making, coordination, and control, information systems may also help managers and workers analyze problems, visualize complex subjects, and create new products. An Information System (IS) is a broader concept that refers to a coordinated set of components designed to collect, process, store, and disseminate information to support decision-making, coordination, control, analysis, and visualization within an organization. Information systems combine IT with people and business processes to create a system that supports and enhances business activities.

Management Information Systems

A Management Information System (MIS) is a computer-based system that collects, processes, stores, and disseminates information to support decision-making, coordination, control, analysis, and visualization within an organization. It gives managers at every level, strategic, tactical, and operational, the right information, in the right form, at the right time, to run and control the business effectively.

What are the Functions of an Information Systems?

An information system has **three basic functions**, plus a fourth one that keeps it accurate. The diagram shows exactly how they work together inside an organization, which sits inside a larger environment.

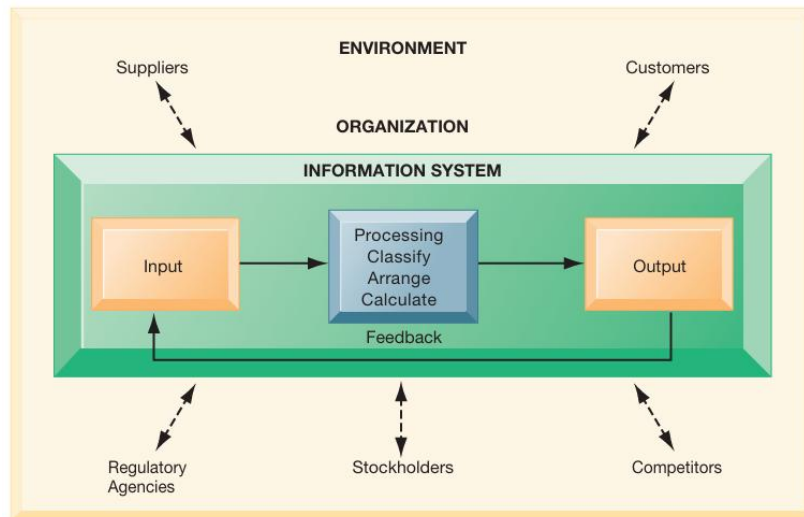
1. Input: This is where raw data enters the system. It could be customer orders, employee attendance, sales transactions, or supplier details. Input is simply data collection; nothing is analyzed yet.

2. Processing: This is where the real work happens. The system takes the raw input and:

- **Classifies** it (sorts data into categories, like grouping sales by region)
- **Arranges** it (organizes it in a logical order, like by date or department)
- **Calculates** it (does the math, like totals, averages, or percentages)

Processing is what turns messy raw data into something useful.

3. Output: This is the finished result, delivered as reports, summaries, charts, or documents that people can actually use to make decisions. For example, a monthly sales report or a payroll summary.



4. Feedback: Notice the arrow looping back from Output to Input in the diagram. Feedback means some of the output is sent back into the system to check and correct the input. For example, if a report shows an unusually high number of sales for one product, someone might go back and check whether the input data was entered correctly.

What is the Dimension of Information System?

To understand an Information System (IS) properly, we should not think of it as simply a computer or a piece of software. An Information System is influenced by three major dimensions



These three dimensions work together to create an effective Information System.

1. Organization

Organization is an important dimension of an Information System. It includes the people, organizational structure, business processes, politics, and culture of an organization. The organizational structure defines different levels of authority and responsibility. Business processes explain how tasks and activities are performed. Organizational culture represents the shared values, beliefs, and ways of doing things. Information Systems support and automate many of these organizational activities and processes.

2. Management:

Management is an important dimension of an Information System. It involves making decisions, solving organizational problems, and developing strategies to respond to business challenges. Managers identify problems and allocate human and financial resources to achieve organizational goals. They also provide leadership and coordinate organizational activities. Information technology helps managers create new products and services and improve existing business processes.

3. Information Technology

Technology is the third major dimension of an Information System. It includes computer hardware, computer software, data management technology, and networking and telecommunications technology. Hardware provides the physical equipment for input, processing, output, and storage. Software provides instructions that control computer hardware. Data management technology organizes and manages data, while networking technology connects devices and allows data to be shared.

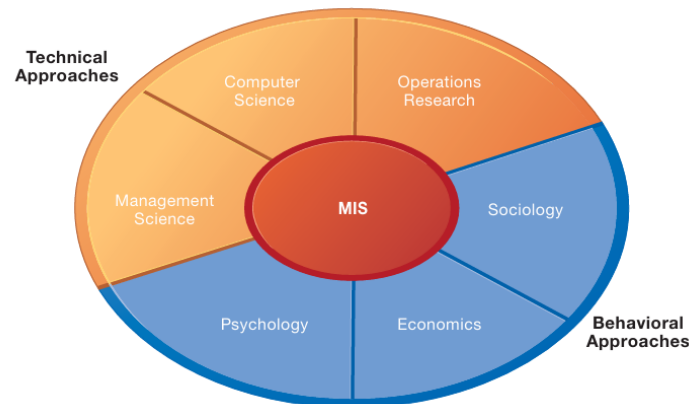
Six Strategic Business Objectives of Information Systems

Every information system a firm build should serve at least one of these objectives.

Objective	What It Means	Example
Operational Excellence	Improving efficiency and productivity to raise profitability.	Walmart's Retail Link system links suppliers directly to store shelf data.
New Products, Services & Business Models	Using IT to create offerings that did not exist before.	Apple's iTunes/App Store replaced the old model of selling physical music/media.
Customer & Supplier Intimacy	Knowing customers/suppliers well enough to serve them better and build loyalty.	Hotels adjusting room settings automatically from a guest's digital profile.
Improved Decision Making	Giving managers real-time, accurate data instead of guesswork.	HR systems that let managers instantly review performance history before a promotion decision.
Competitive Advantage	Combining the objectives above to outperform rivals.	Firms that consistently do the above out-earn competitors who do not.
Survival	Meeting basic requirements to keep operating in an industry.	Banks must offer ATM/network access; firms must retain digital records to meet legal requirements.

What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

Information Systems (IS) is multidisciplinary, meaning it draws knowledge from several academic disciplines. It is generally divided into Technical and Behavioral approaches. Information Systems are also considered sociotechnical systems, combining technology with social and organizational factors.



Technical Approach

The technical approach focuses on the technology and mathematical models of Information Systems. It includes:

- **Computer Science** - studies computing, data storage, and access.
- **Management Science** - develops models for decision-making and management.
- **Operations Research** - uses mathematical techniques to optimize business activities such as transportation and inventory.

Behavioral Approach

The behavioral approach focuses on people, organizations, and behavior related to Information Systems. It includes:

- **Sociology** - studies how organizations and groups influence IS and how IS affects them.
- **Psychology** - studies how people perceive and use information.
- **Economics** - studies digital goods, digital markets, and the impact of IS on business costs and control.

Review Question

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